Such a potential can be fitted to the basin of a bond energy function. The molecular motion should be confined to the bottom of the basin for such a simplified harmonic model to be realistic. Of particular importance in models of organic matter are the bonds to hydrogen atoms CH, OH, NH, etc., which tend to have very large harmonic coefficients and therefore give rise to the highest frequency vibrations in the system. Other types of bonds include *ionic bonds* (in which one atom donates an electron to another atom creating a charge imbalance) and *hydrogen bonds*, which are relatively weak interactions that arise as polarization effects between hydrogens and other atoms.

Another type of force arises in multi-electron atoms due to the fluctuations of the charge fields around the atoms. This leads to an instantaneous polarization and an attractive interaction termed *London dispersion*; it is most often modelled using an inverse sixth power potential:

$$\varphi_{\text{disp}}(r) \sim -\frac{K}{r^6}, \qquad K > 0.$$

On the other hand, as the atoms come together *Pauli repulsion* comes into play. As the nuclei approach one another, the wave-functions of the electrons in the outer shells of the two atoms will have significant overlap. Due to the Pauli exclusion principle, some of the electrons in these shells are forced into higher energy states and this leads to a higher overall energy (which grows as the internuclear distance is decreased); this is felt as a strong net repulsive force. There are many models for this repulsive wall. The tension between attraction and repulsion creates an energy well. This behavior is incorporated in most smooth short-ranged potential energy models. The combination of dispersive and repulsive forces are referred to as the *van der Waals forces*. Because no electrons are shared between the atoms, the van der Waals forces are much weaker than covalent bonds (in the sense that the potential energy well created due to van der Waals interaction is much more shallow than that associated to a bond).

Buckingham [56] suggested a combined potential of the form

$$\varphi_{\text{B}}(r) = Ae^{-Br} - \frac{C}{r^6}, \qquad A > 0,\ B > 0,\ C > 0.$$

A more common choice in simulation is the Lennard-Jones (6–12) potential

$$\varphi_{\text{LJ}}(r) = 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right].$$

The primary motivation for use of Lennard-Jones potentials in simulation is the fact that the 12th power can be obtained by a simple squaring of the 6th power, making the computation of the energy (and forces) relatively fast, however the efficiency obtained from this is not necessarily important, depending on the application and the type of computer hardware available. Because of the strongly repulsive character

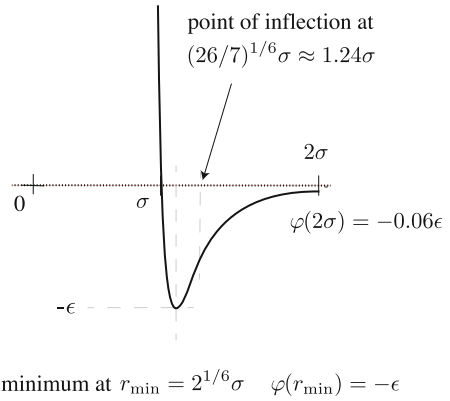


Fig. 1.6 The Lennard-Jones pair potential $\varphi(r) = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$ is shown

of short-ranged soft walls in molecular dynamics, i.e., the fact that φ_{LJ} tends rapidly to positive infinity as $r \to 0$, the atoms remain well separated in long simulations. The singularity at $r = 0$ is therefore rarely encountered in dynamics trajectories, however the presence of the singularity may nonetheless create problems for mathematical analysis, as many theoretical techniques rely on assumed smoothness. An alternative would be to replace the Lennard-Jones potential with a smooth fitted (e.g. piecewise polynomial) potential that captures, for a sufficiently large range of energies, both the attractive behavior and the strong repulsion, while mimicking the Lennard-Jones potential adequately on a sufficiently large interval, without introducing a singularity.

When there are many atoms of different types, the parameters of the potential will depend on this, so we obtain contributions to the total potential energy of the form

$$\varphi_{ij}^{\text{LJ}}(r_{ij}) = 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right],$$

where $r_{ij} = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|$. The Lennard-Jones potential is graphed in Fig. 1.6.

1.1.2 Other Potentials

As we have already mentioned, an important and growing application area for molecular simulation methods lies in the modelling of biological molecules. For example, the alanine dipeptide is a very popular model for examining different simulation algorithms intended for use in biomolecular modelling. It consists of 22 atoms (carbon, oxygen, hydrogen, nitrogen) arranged as shown in the ball and stick diagram of Fig. 1.7.